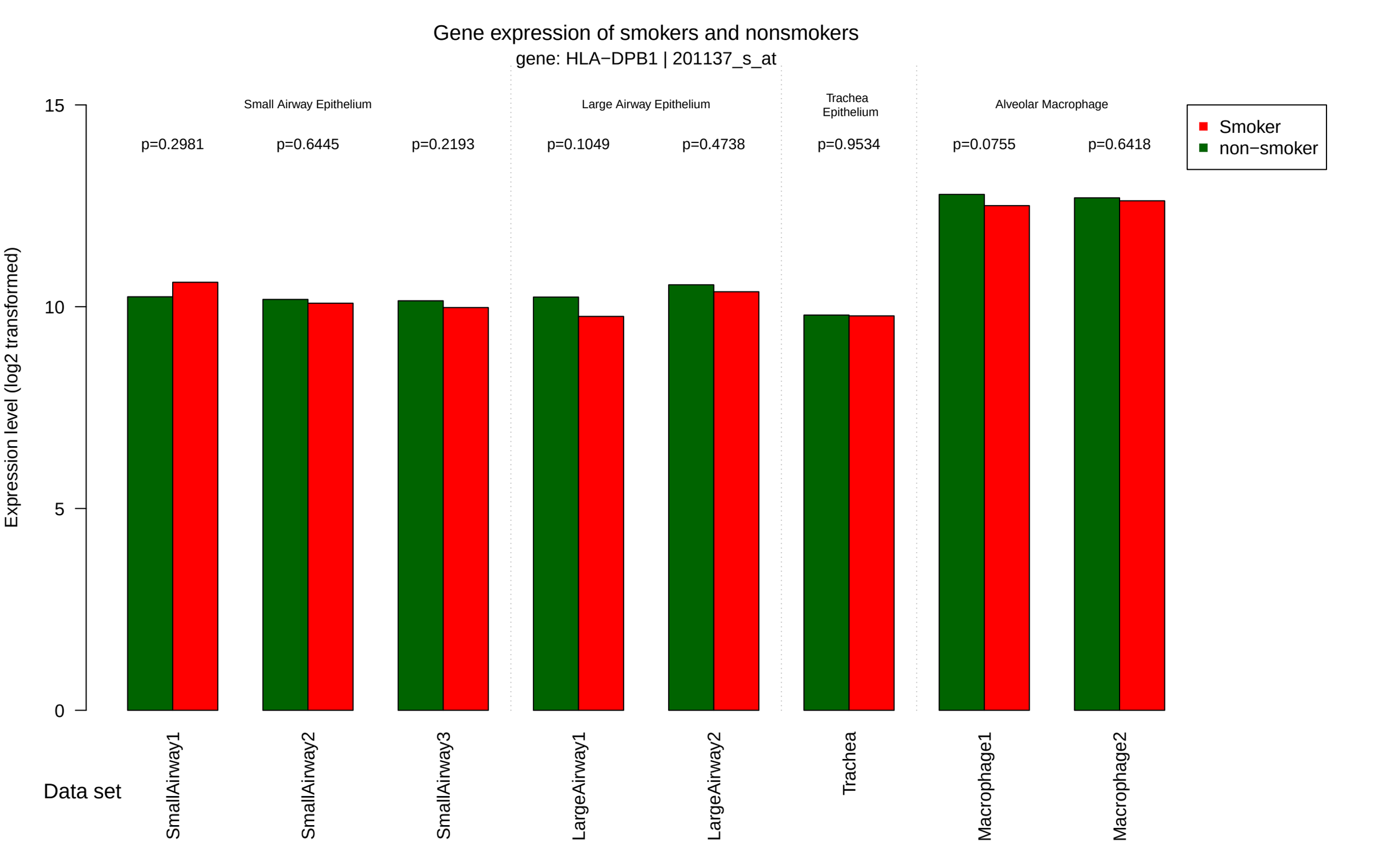
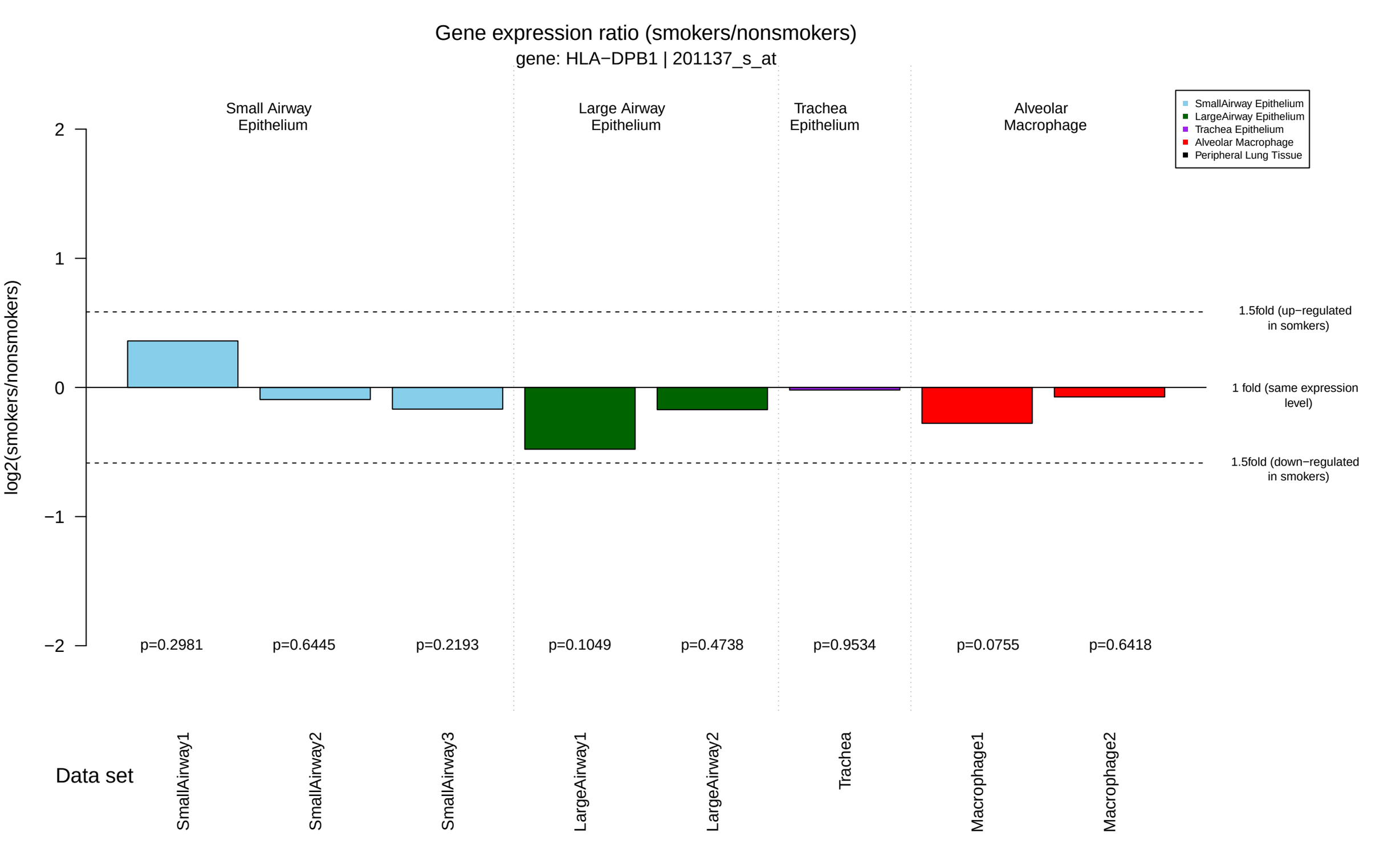
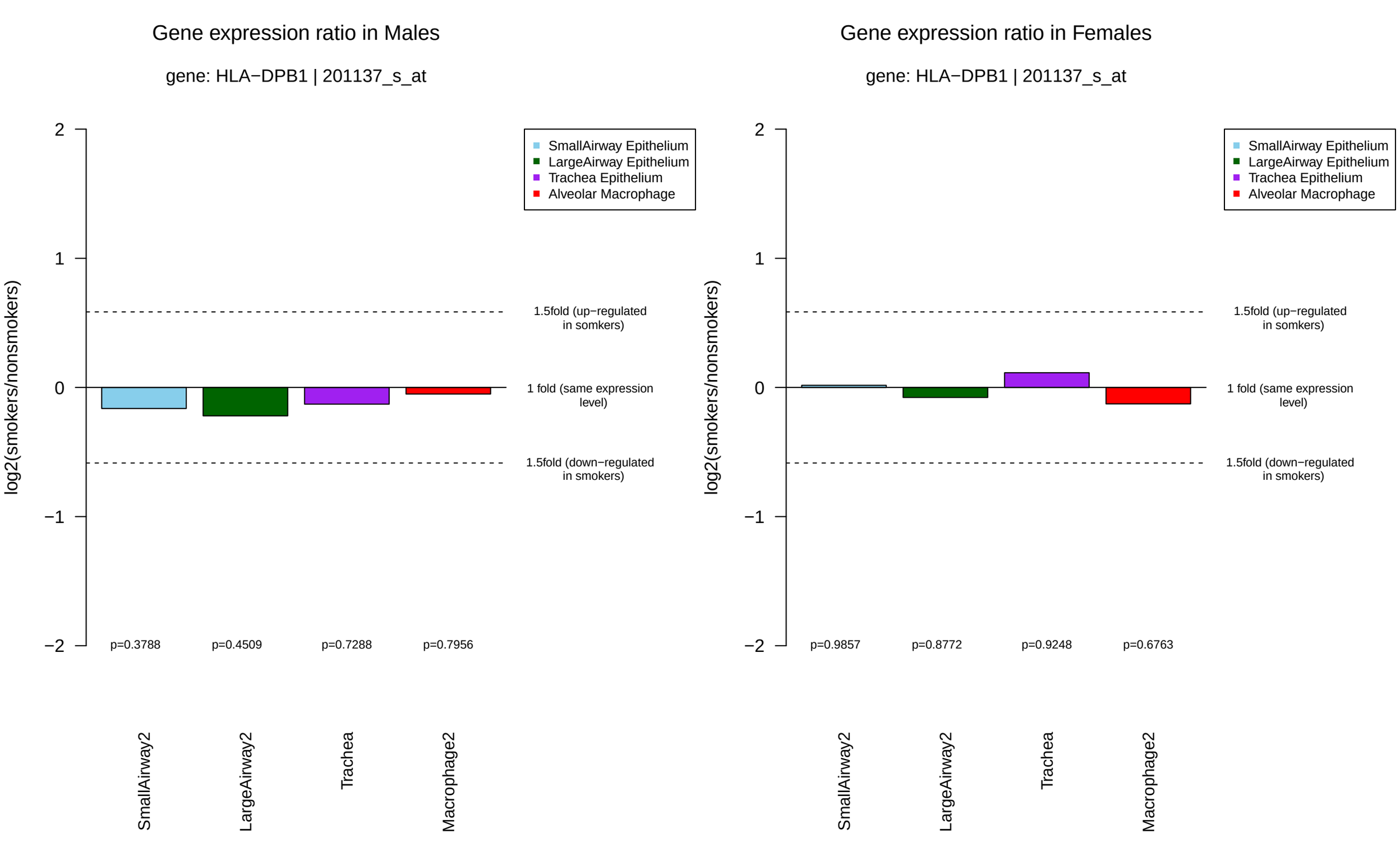
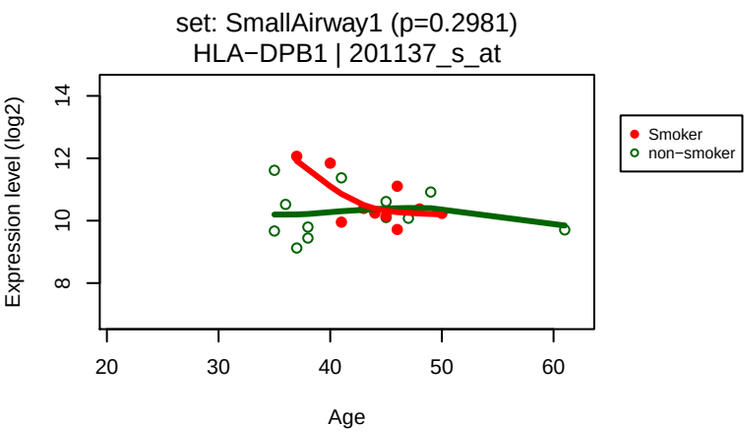
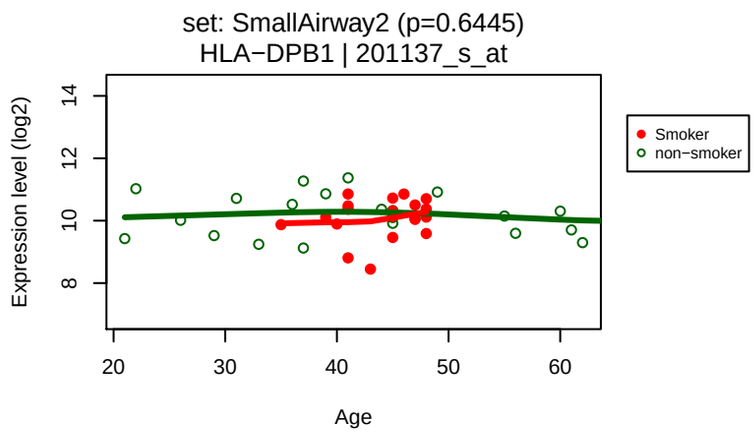
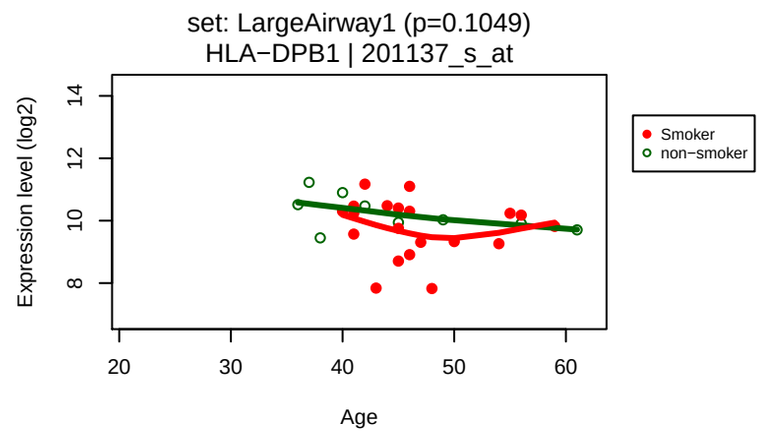

Gene expression of smokers and nonsmokers
gene: HLA−DPB1 | 201137_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2981
p=0.6445
p=0.2193
p=0.1049
p=0.4738
p=0.9534
p=0.0755
p=0.6418
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

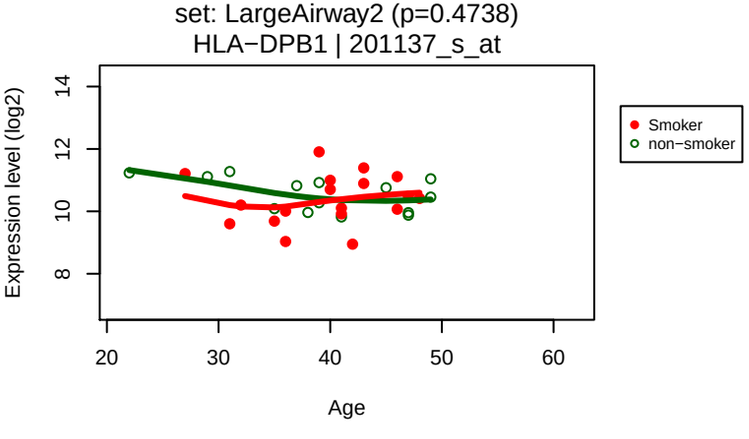

Gene expression ratio (smokers/nonsmokers)
gene: HLA−DPB1 | 201137_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2981
p=0.6445
p=0.2193
p=0.1049
p=0.4738
p=0.9534
p=0.0755
p=0.6418
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

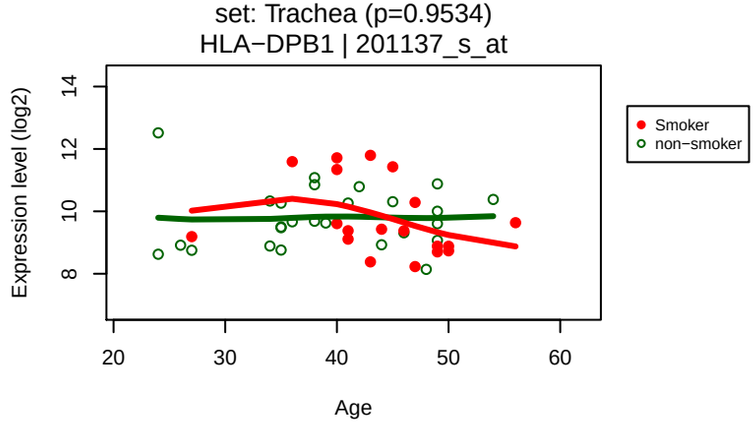

Gene expression ratio in Males
gene: HLA−DPB1 | 201137_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3788
p=0.4509
p=0.7288
p=0.7956
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HLA−DPB1 | 201137_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9857
p=0.8772
p=0.9248
p=0.6763
SmallAirway2
LargeAirway2
Trachea
Macrophage2

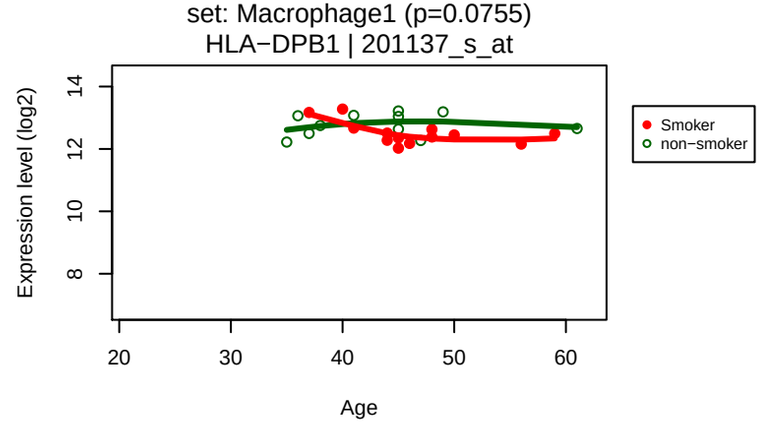

set: SmallAirway1 (p=0.2981)
HLA−DPB1 | 201137_s_at
Expression level (log2)
Age
Smoker
non−smoker

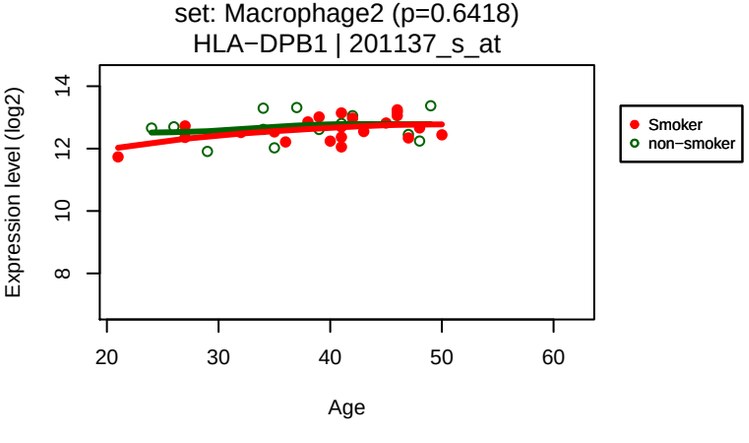

set: SmallAirway2 (p=0.6445)
HLA−DPB1 | 201137_s_at
Expression level (log2)
Age
Smoker
non−smoker

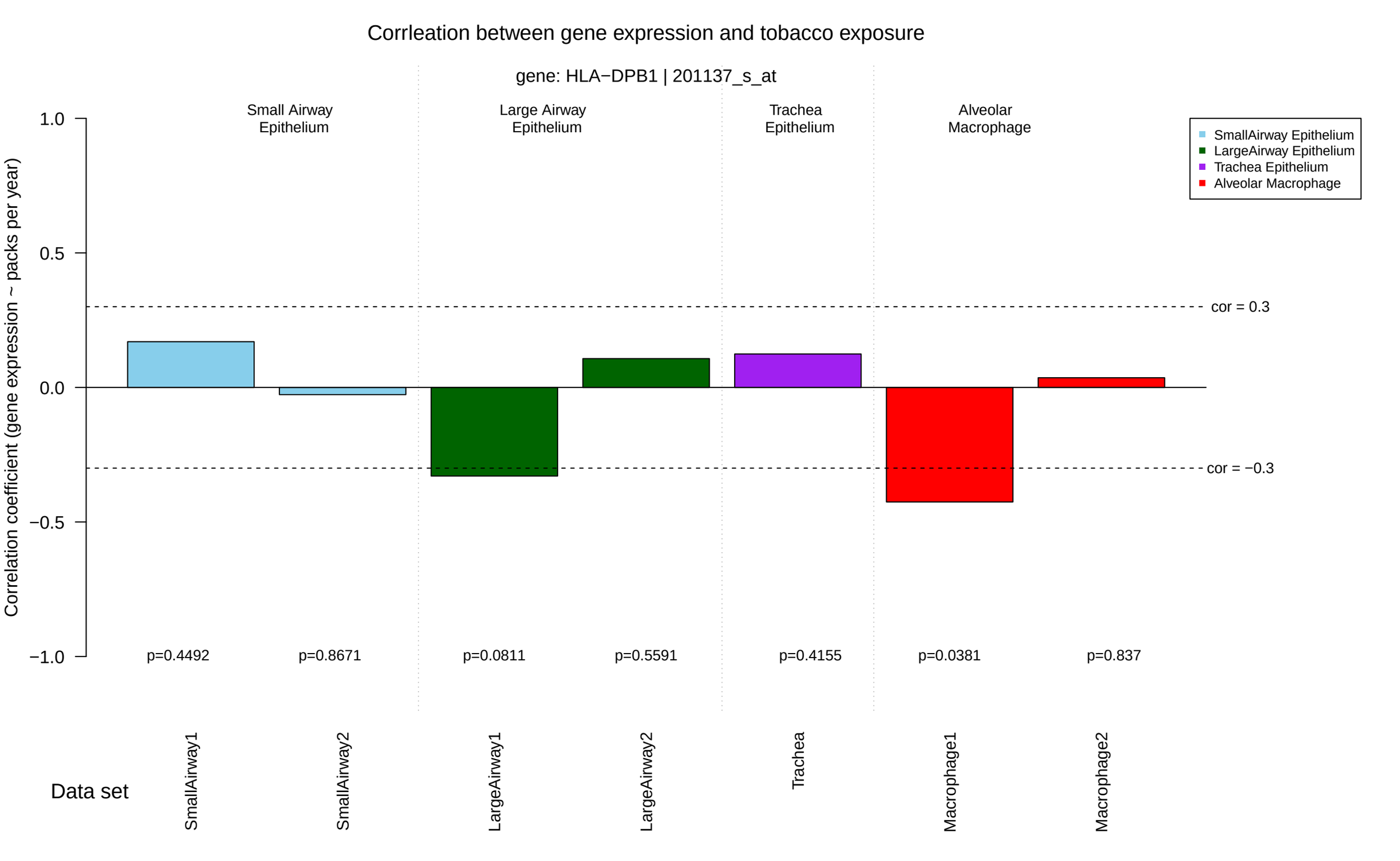

set: LargeAirway1 (p=0.1049)
HLA−DPB1 | 201137_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4738)
HLA−DPB1 | 201137_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.9534)
HLA−DPB1 | 201137_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0755)
HLA−DPB1 | 201137_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6418)
HLA−DPB1 | 201137_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HLA−DPB1 | 201137_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4492
p=0.8671
p=0.0811
p=0.5591
p=0.4155
p=0.0381
p=0.837
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2